User Interface design

Version 3.0

Or, Usability

(NF requirement)

For more

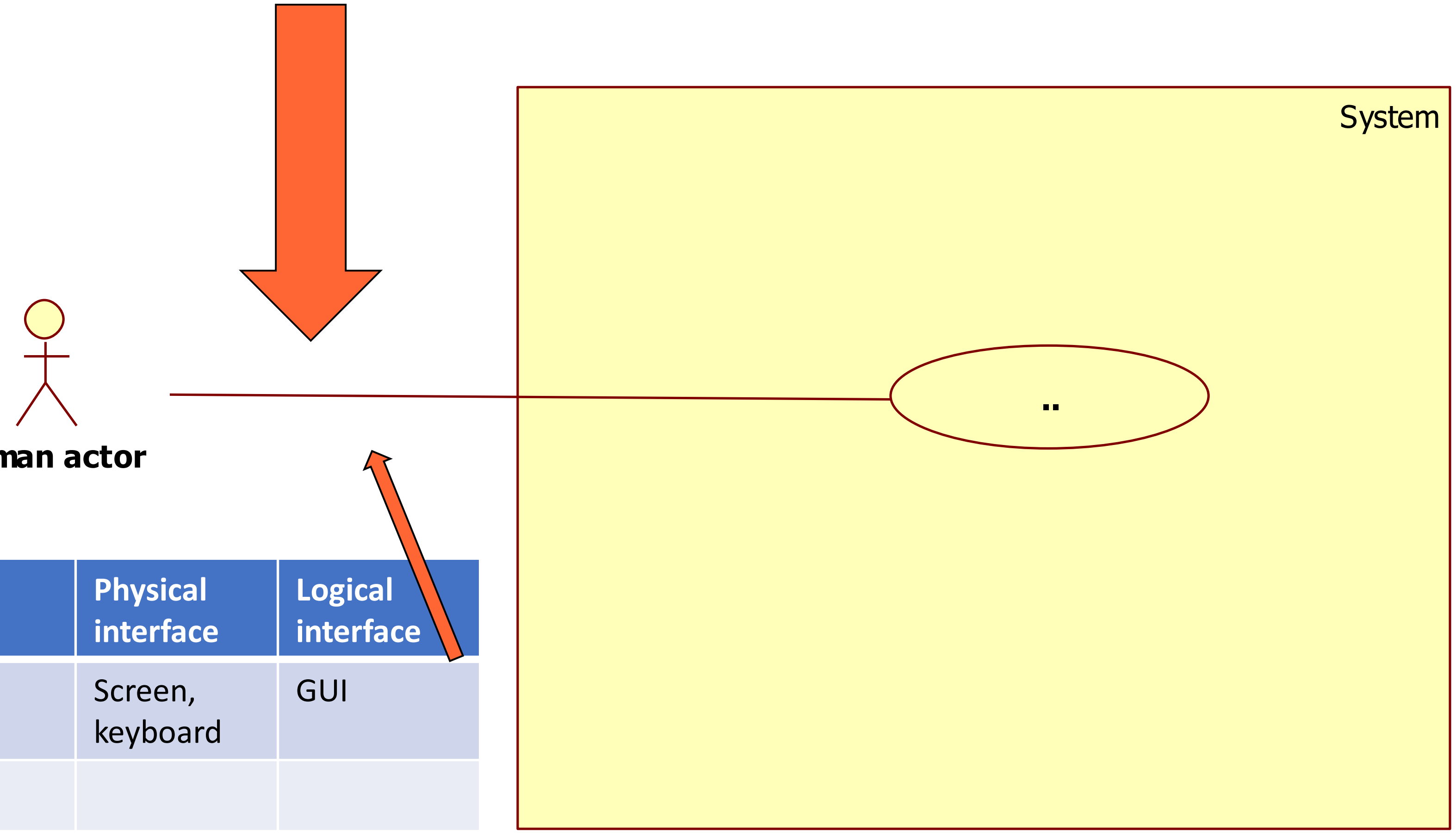
- 02JSKOV Human Computer Interaction
- Prof. Luigi De Russis

UI Design

- When human actors are involved, designing the User Interface (often Graphical User Interface) is a key design choice
- We assume that RE activity has been completed
  - (in practice RE and UI design may overlap)

Example

(G) UI



Actor	Physical interface	Logical interface
Human actor	Screen, keyboard	GUI
..		

Key message #1

- Experimentation

- The UI should be experimented with the end users

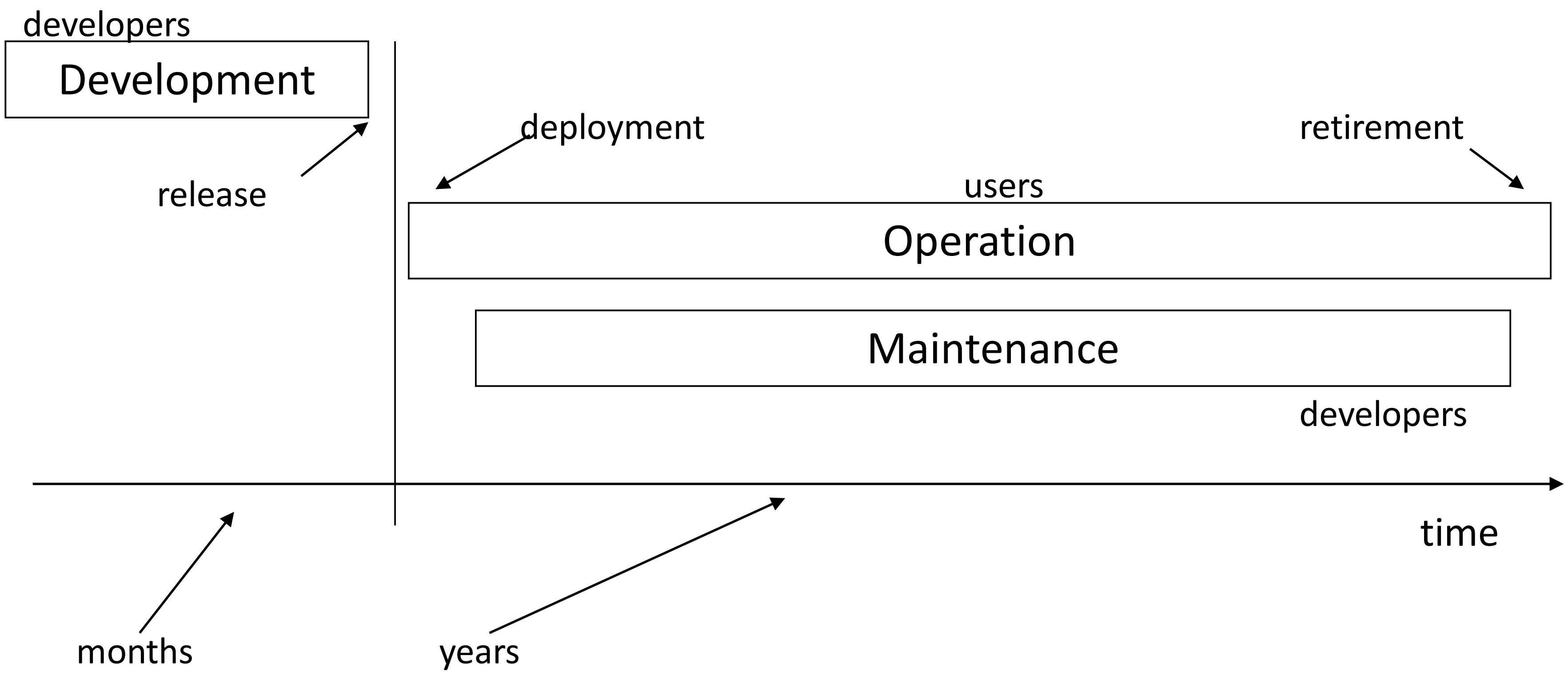
Key message #2

> "A user interface is like a joke. If you have to explain it, it's not that good."
>
> MARTIN LEBLANC

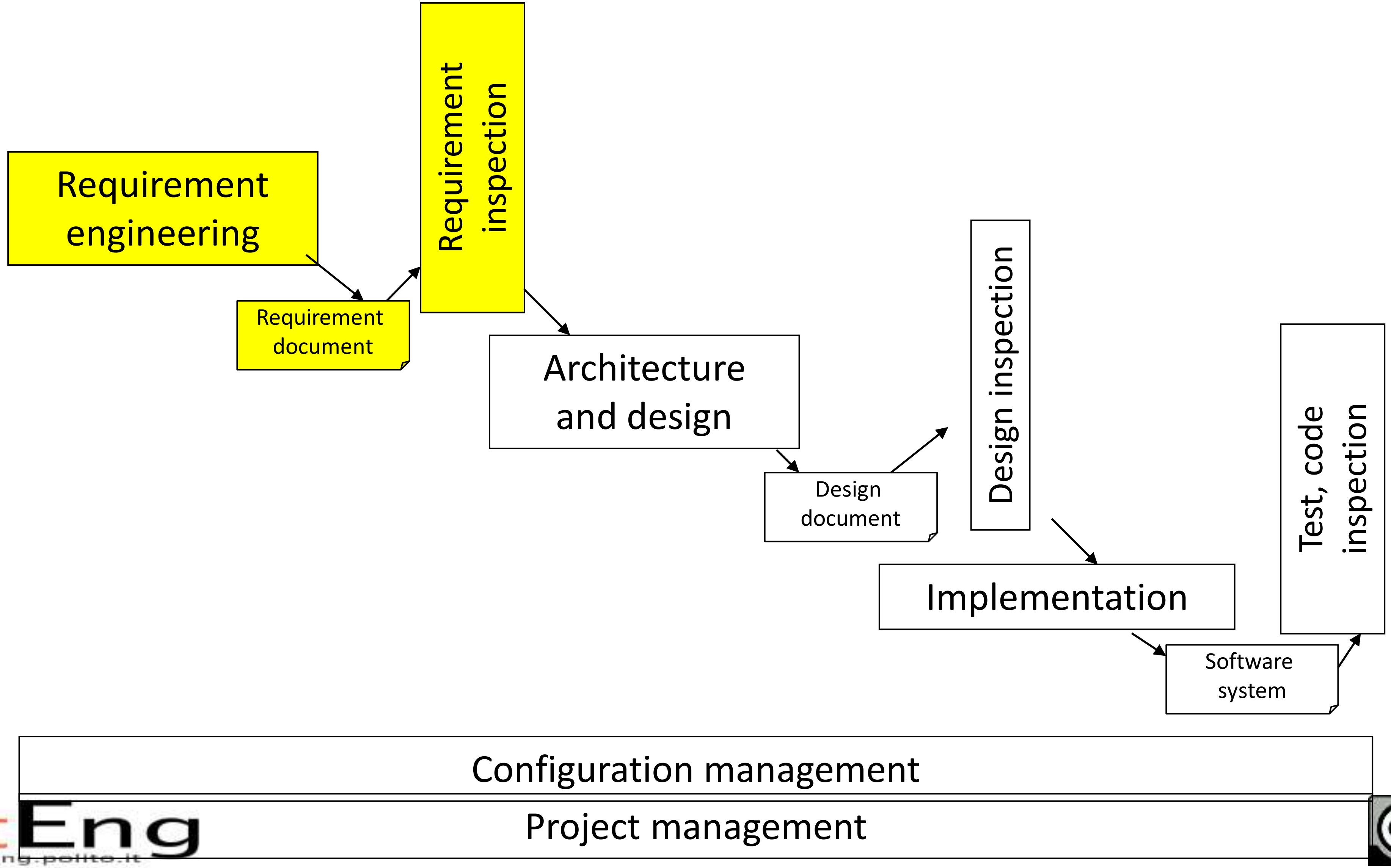
How to

- Starting points are
  - Context diagrams, actors
  - Functional requirements
  - Use cases

The process - phases



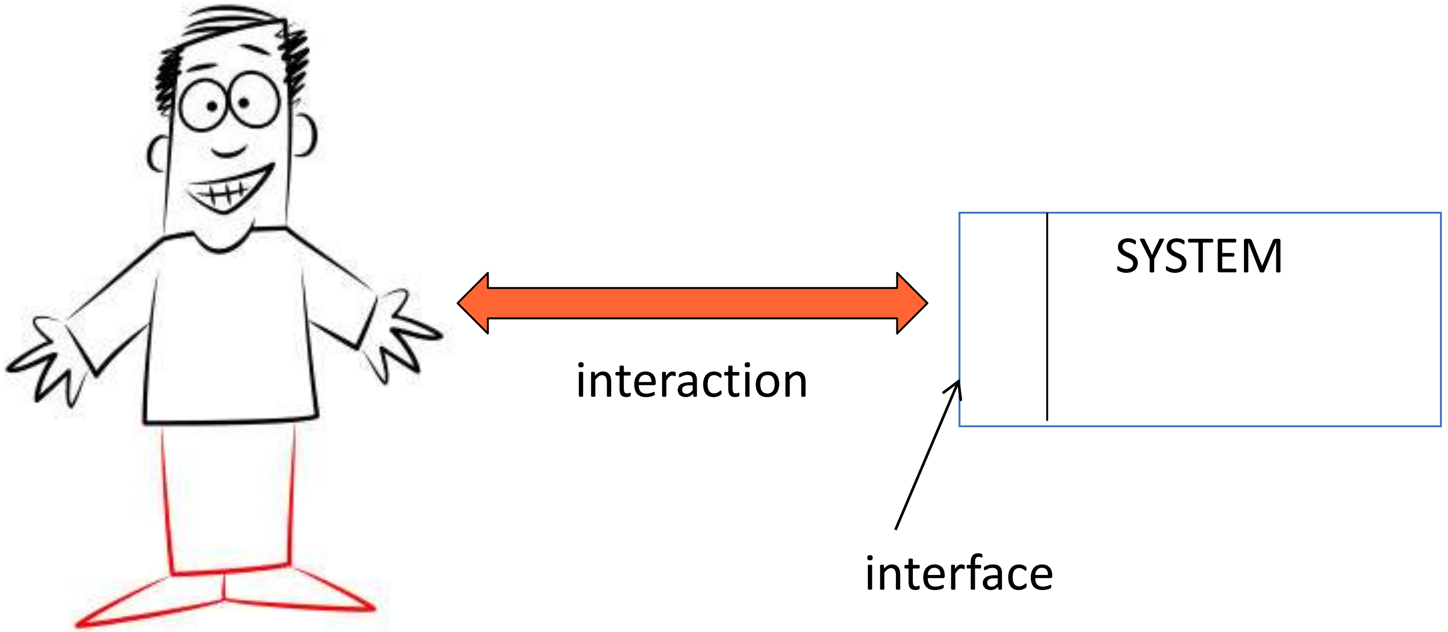
The process - development



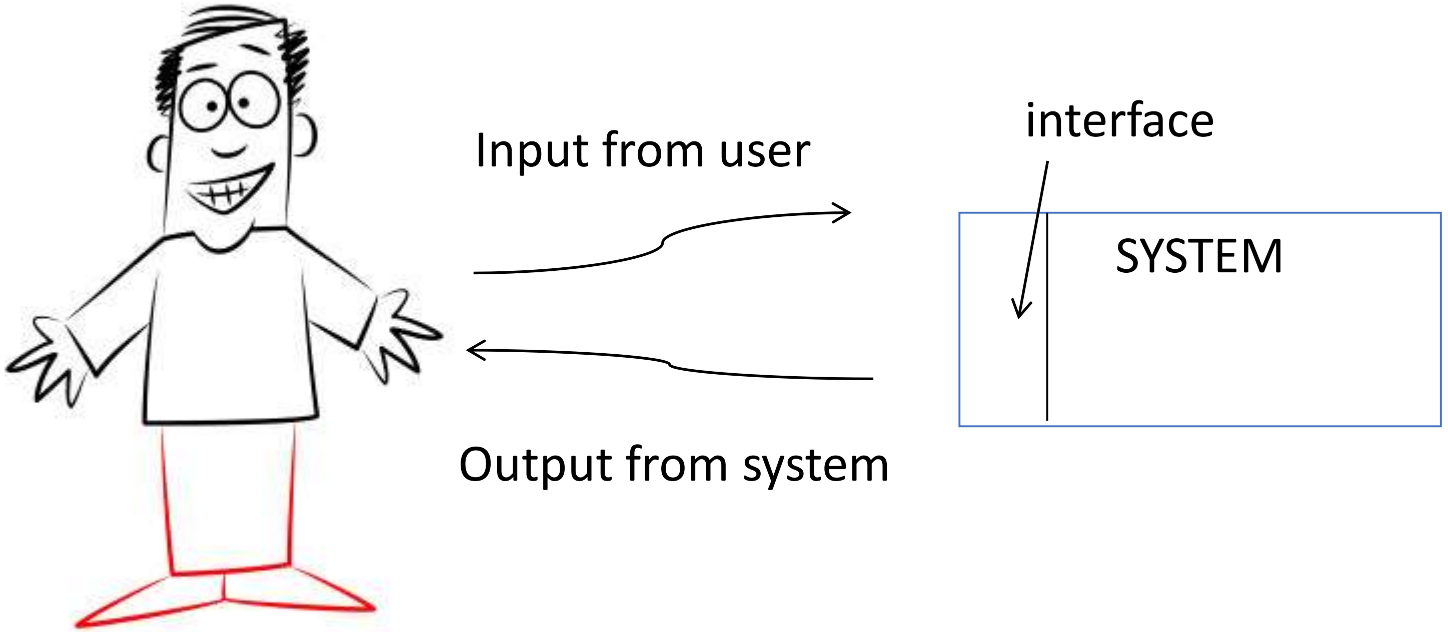
Context

- Software types
  - Embedded software
  - Mass market / consumer software
  - Enterprise software

Context



Interaction

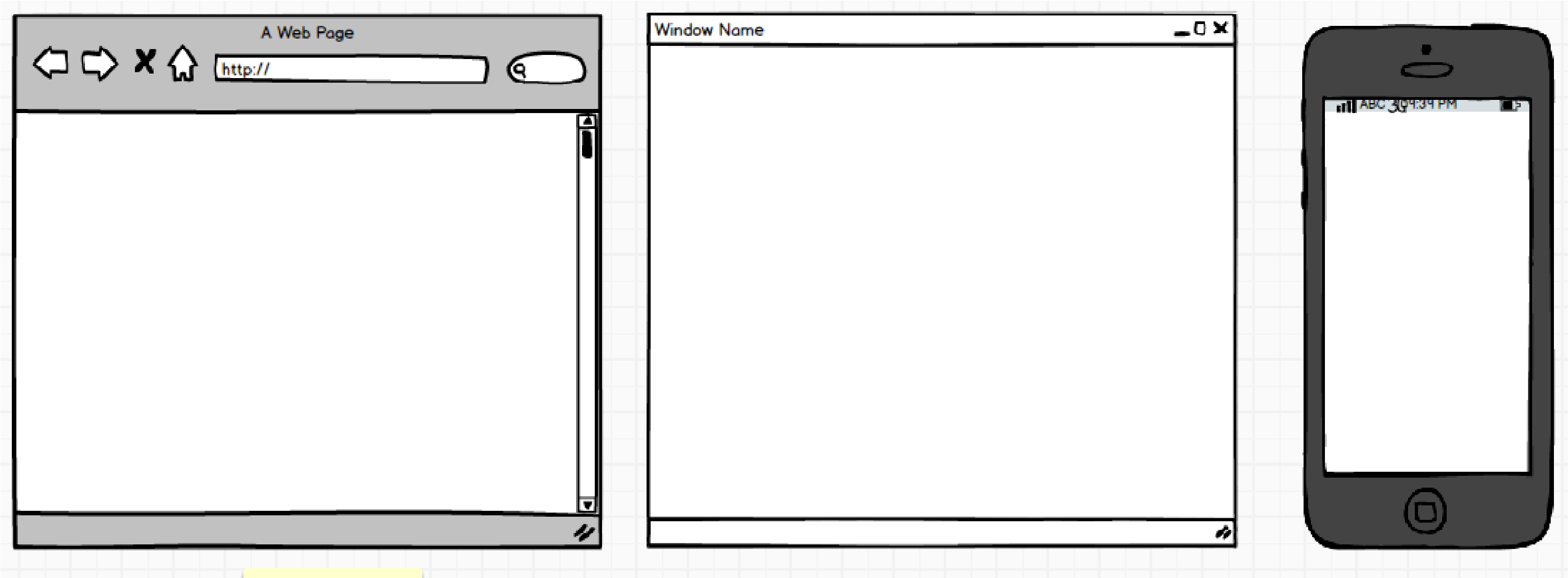


Interaction means

User		System
Sight		Screen, printer, glasses.
Hearing		Noise, music, voice synthesis
Touch		Glove
Hands		Keyboard, mouse, touchscreen, touchpad, glove
Voice		Voice recognition
Eyes		Eye tracking
Position, gesture		Gesture recognition

Context

- One application, many UIs



Web Desktop Mobile

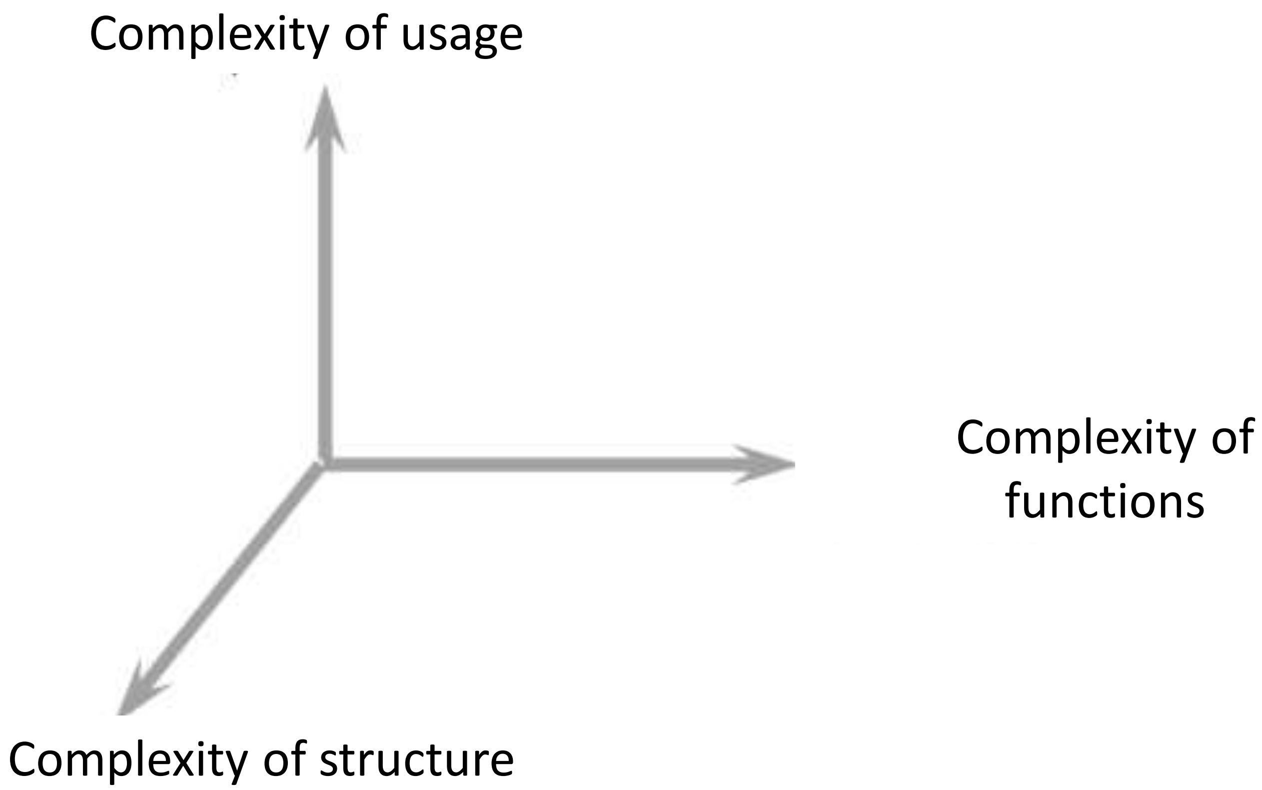
Context

- One application, many UIs

Tablet

Smartwatch

Principles - simplicity



Simplicity

- UI should be as simple, as allowed by complexity of structure, of functionality

Ex RVC - basic

- Functions: clean, (map)
- Interaction
  - Option1: 1 button
    - press 3 seconds: on, press 5 seconds: off, press 3 times quickly: clean
  - Option2: 2 buttons + 4 inch display
    - On off, clean. Display shows status
  - Option3: 3 buttons
    - On off, clean, map
  - Option4: App on smartphone, several windows

Which is best option?

Ex RVC - advanced

- Functions:
  - Manage multiple spaces (many rooms, many houses)
  - Plan cleaning (every Tuesday at 6am do room1, wed at 7 am room 2)
  - Show status (where robot is, bag is full, ..)
  - Show maps collected

- Structure
  - One charging station per space
- UI
  - Smartphone app with several windows and commands

Ex RVC

- Basic vs advanced
  - More functions, more structure, more complex UI

- Basic
  - Given set of functions (ex basic version) UI can be more or less simple

Keep it simple!!

Personas

- Identify and describe typical users
- A persona is a subset of actor (see later)

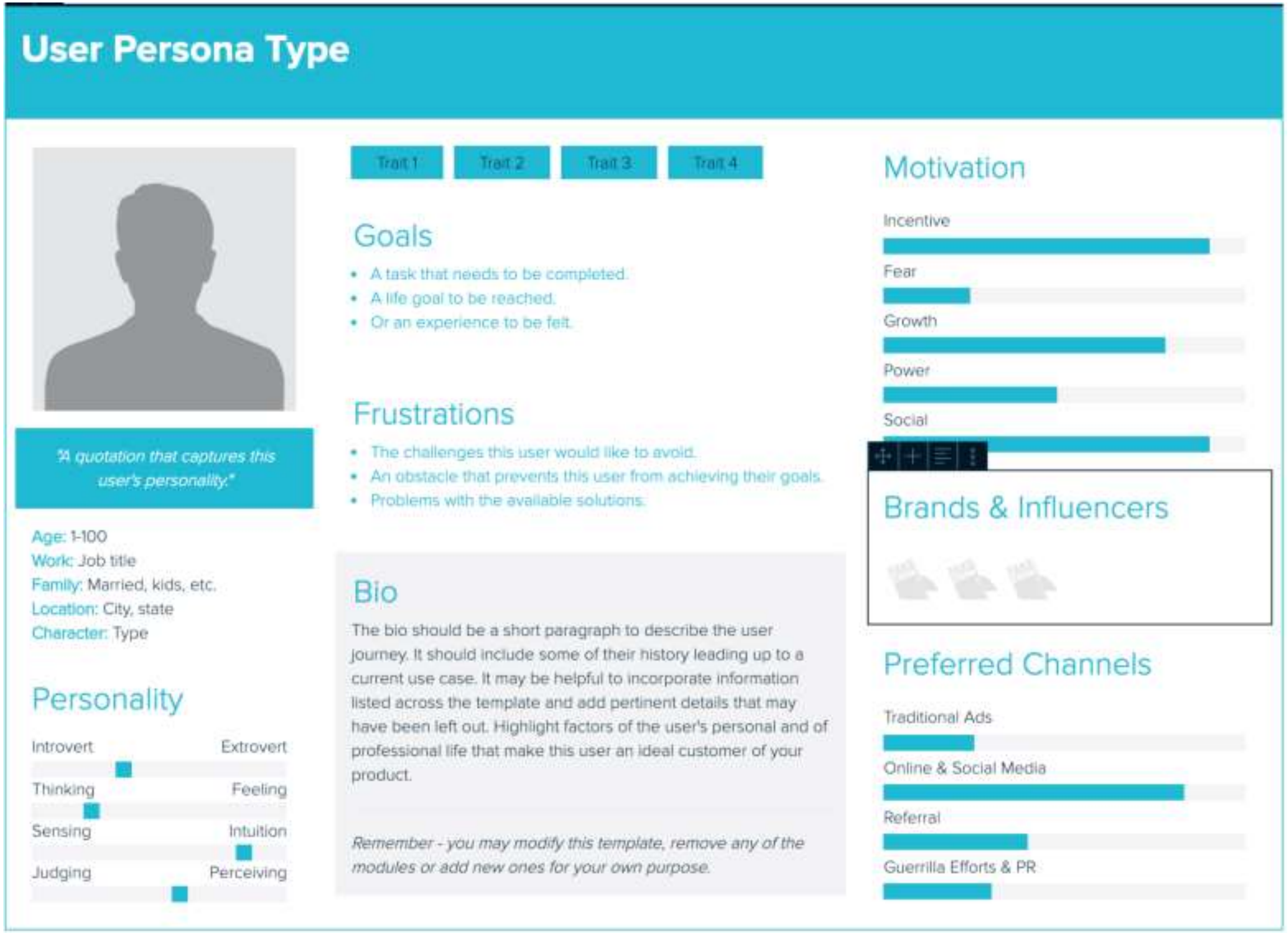
  - Ex persona1: female, middle age, professional, high income, married with children
  - Ex persona2: male, young, student, low income, not married

Personas

- For each persona describe life scenarios
  - Persona1, work day: wake up, breakfast, drive children to school, drive to office ...
  - Persona1, week end day: ...

- For each persona /scenario identify possible interaction with application/object

Example



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Stories

- Help to understand interactions

- Are a cheap way to illustrate design solution from user's (persona's) point of view

- Tell user's goals, motivations and actions

Stories

- "What should this product do?"
- "How would user behave in this context?"
- "What if...?"

Stories

- Without your solution, present-based
  - Focus is set on current practices that illustrate ‘state of the art’ and the problem context
- With your solution, future-based
  - Focus on how problems could be addressed (without diving into details and jargon)

Stories

- In what settings will the product be used?
- Is the persona frequently interrupted?
- With what other products will it be used?
- What primary activities does the persona need to perform to meet her goals?
- What is the expected end result of using the product?

Example: EZGas

- Persona 1: high-income professional, male, married, with children, 45 yo
  - Story: need to visit a client, morning in rush hour, is late, need to find closest open station, no matter what price
- Persona 2: doctor, female, 35
  - Story: back from a long hospital shift, 2 am, alone, need to find the safest place to refuel

Example: EZGas - 2

- Persona 3: student, male, 22, low income
  - Story: borrowed car from a friend, need to find the cheapest place to refuel
- Persona 4: young professional female, 30, is very sensitive about pollution and greenhouse emissions, has a hybrid car
  - Story: commutes home every weekend, needs to find a more sustainable fuel provider
- Persona 5: similar to 4, has a fully electric car
  - Story: needs to find a charging station

In summary

- Personas + stories help to find
  - Functionalities to be offered to satisfy categories of needs (profiles / personas)
  - (functionalities NOT to be offered)
  - Interaction modalities
    - Ex mobile usage on car vs desktop usage at home

Approaches

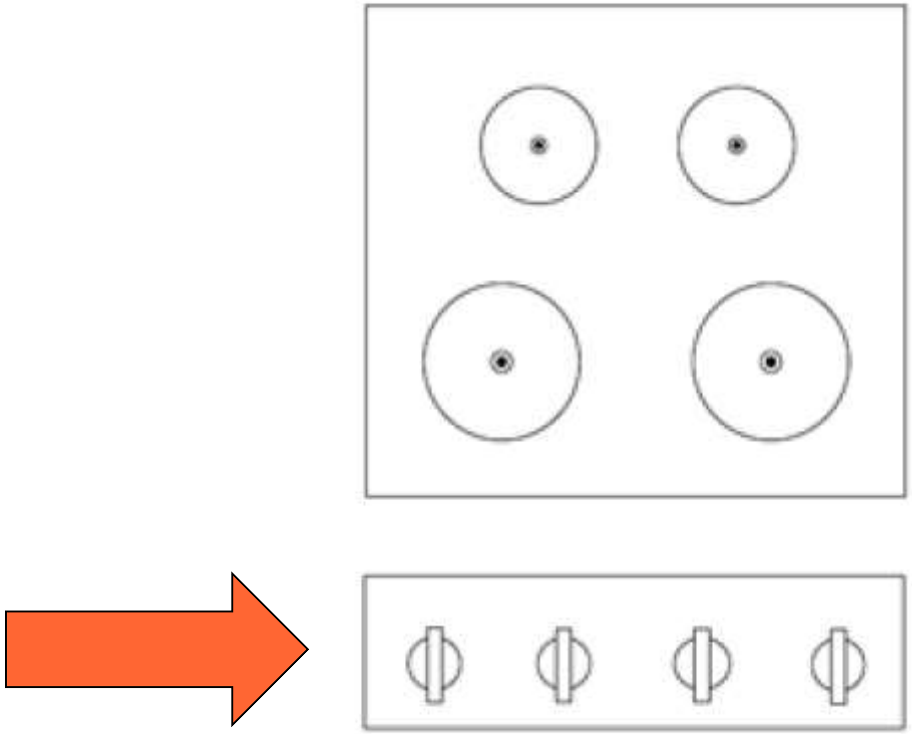
- Ergonomy
  - Safety, adaptability, comfort, usability,...
- Emotional design
  - Beyond ergonomy, the interaction (object) should cause positive emotions in the user
- User eXperience (UX)
  - Usability + feelings + emotions + values

Approaches

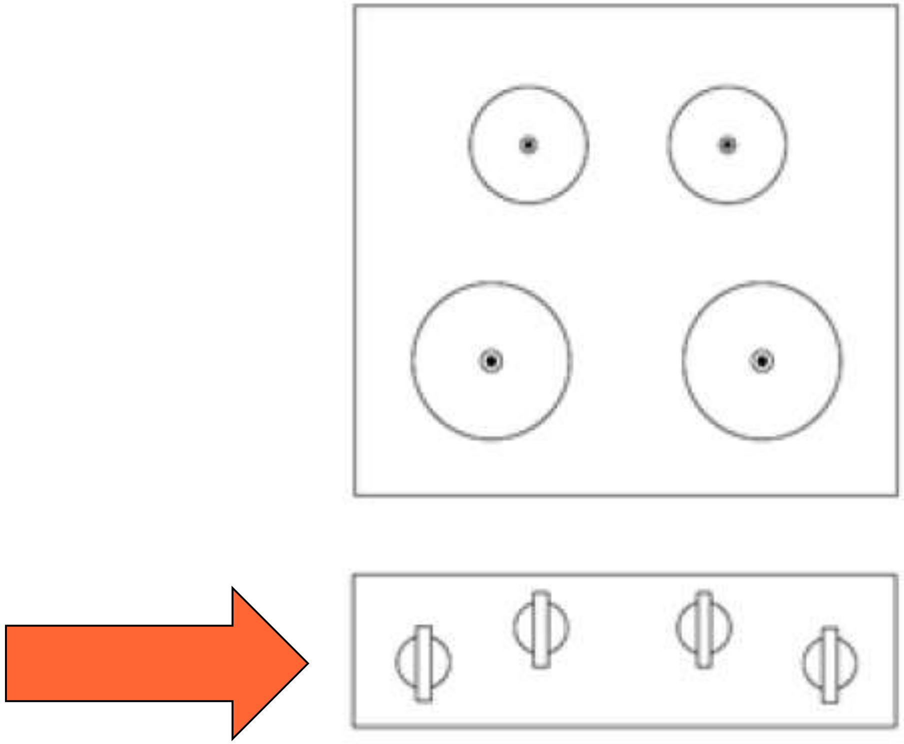
- Transparent technology
  - No emphasis on technology
- Feedback, user centered design
  - No decision based on personal opinions, but feedback from real users

Cooker

Cooker UI



Cooker UI



And, how do i turn the TV on?

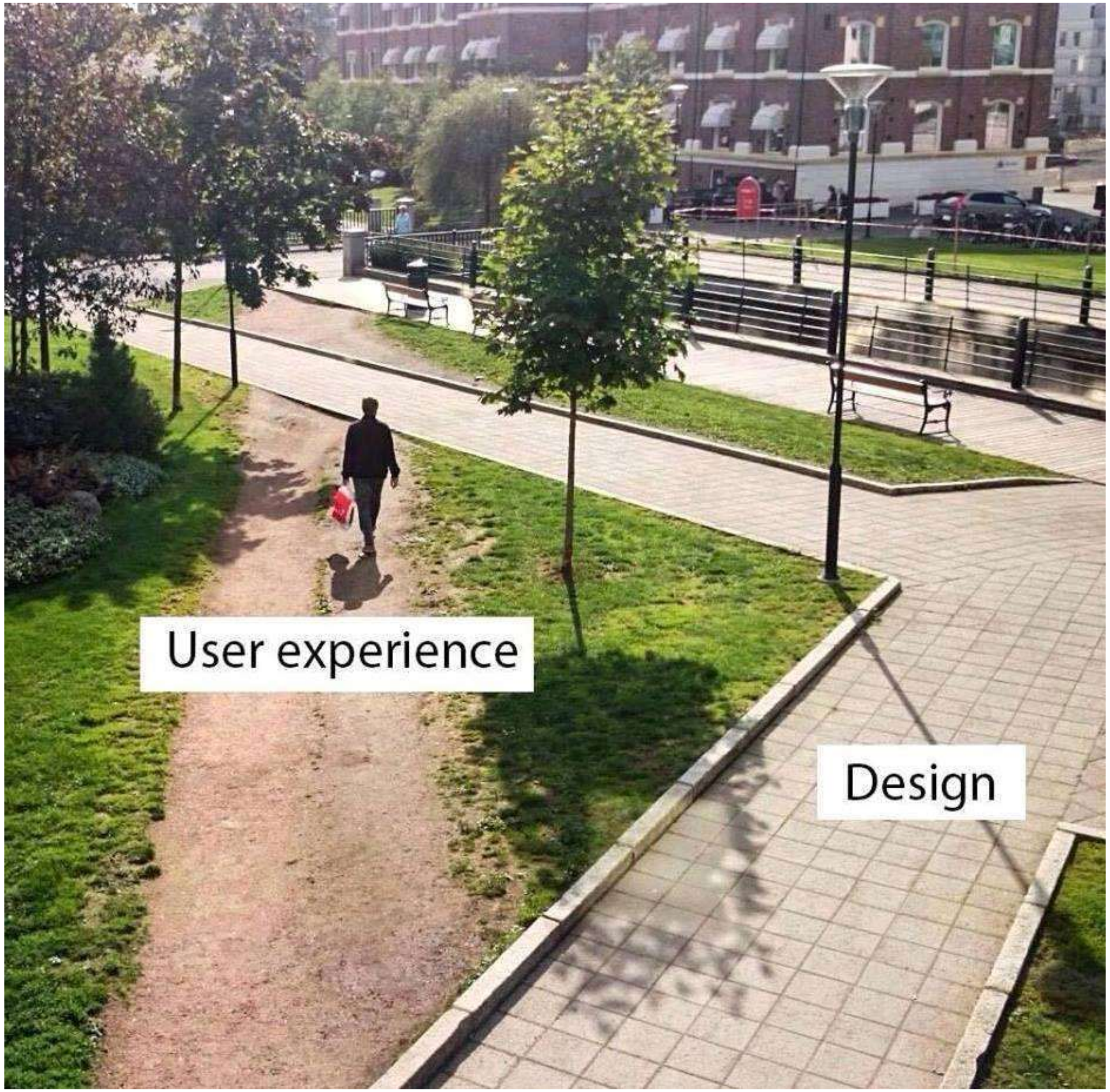
Two screens and two keyboards

One screen

Keyboard + touchscreen

And finally…

ONE SINGLE interaction point



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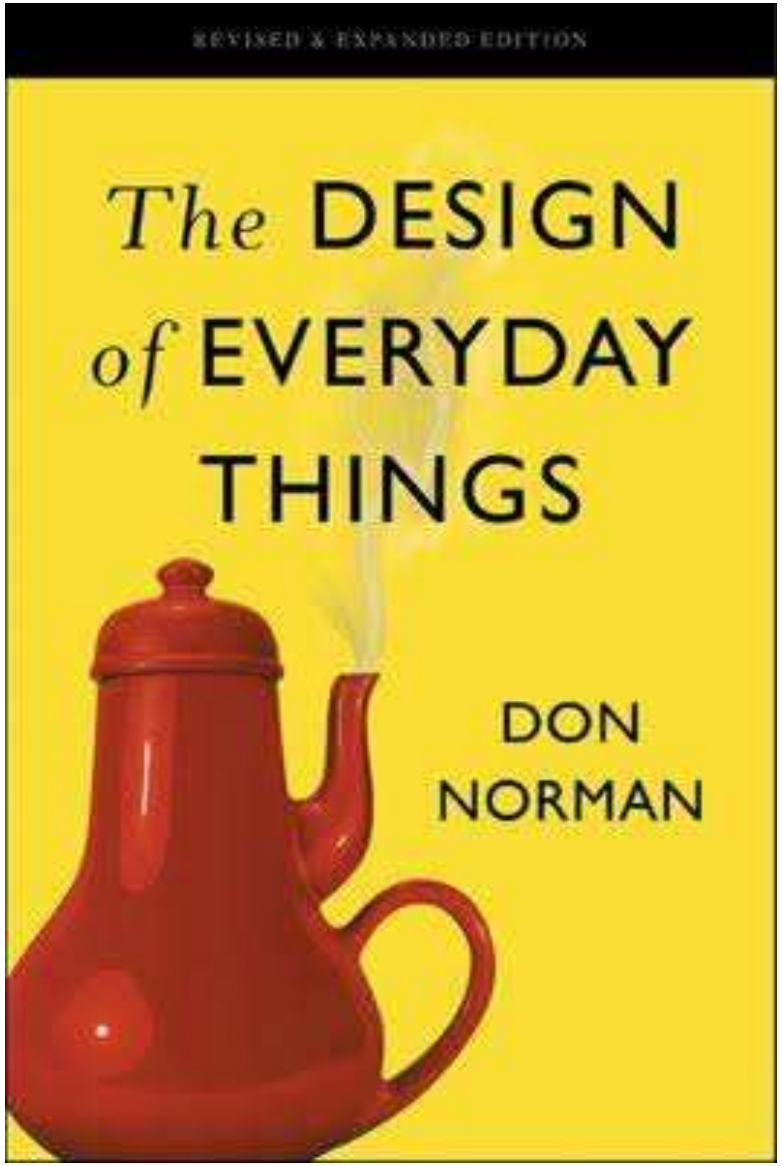
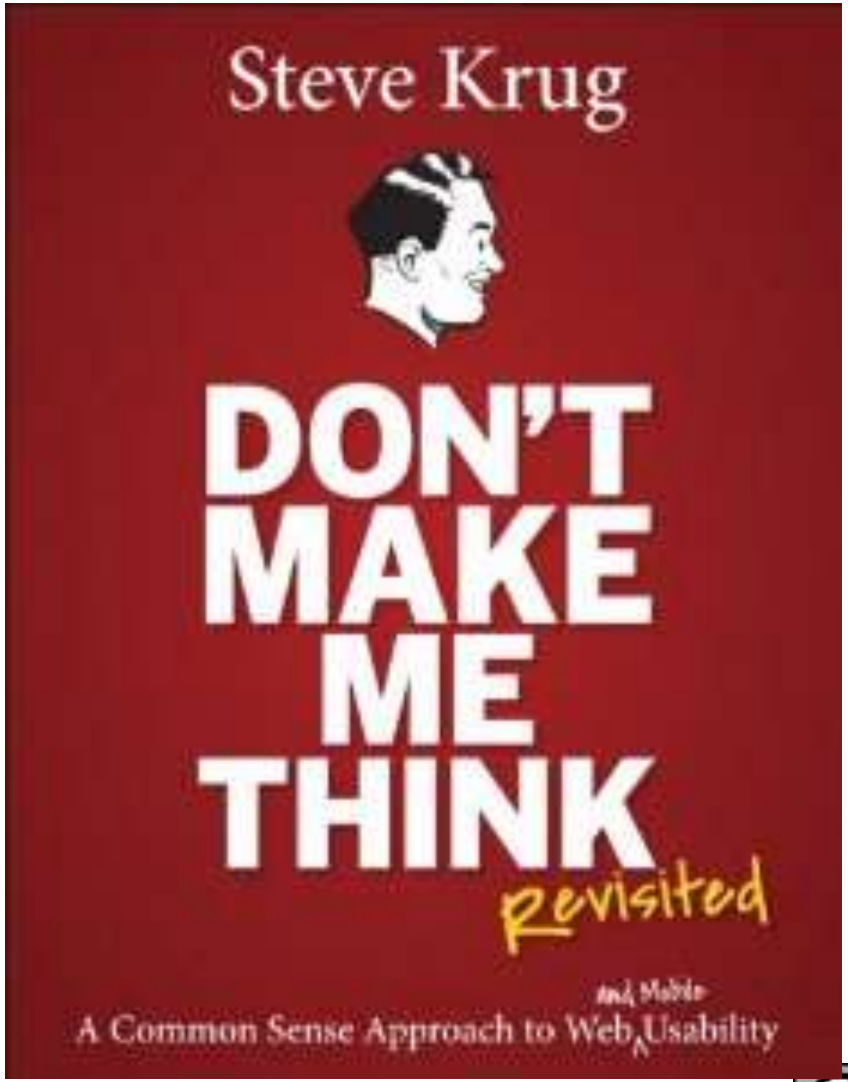

REVISED & EXPANDED EDITION
The DESIGN of EVERYDAY THINGS
DON NORMAN


Steve Krug
DON'T MAKE ME THINK
revisited
A Common Sense Approach to Web and Mobile Usability

User centered design

- Context: mass market product

UCD process - techniques

Activity#	Activity	Techniques
1	Identify the users	Context diagram, personas / actors
2	Define requirements	Use cases, scenarios, functional requirements
3	Define system and interactions	Prototypes
4	In lab tests	Ethnographics, Interviews
5	In field tests	A/B testing Measurements

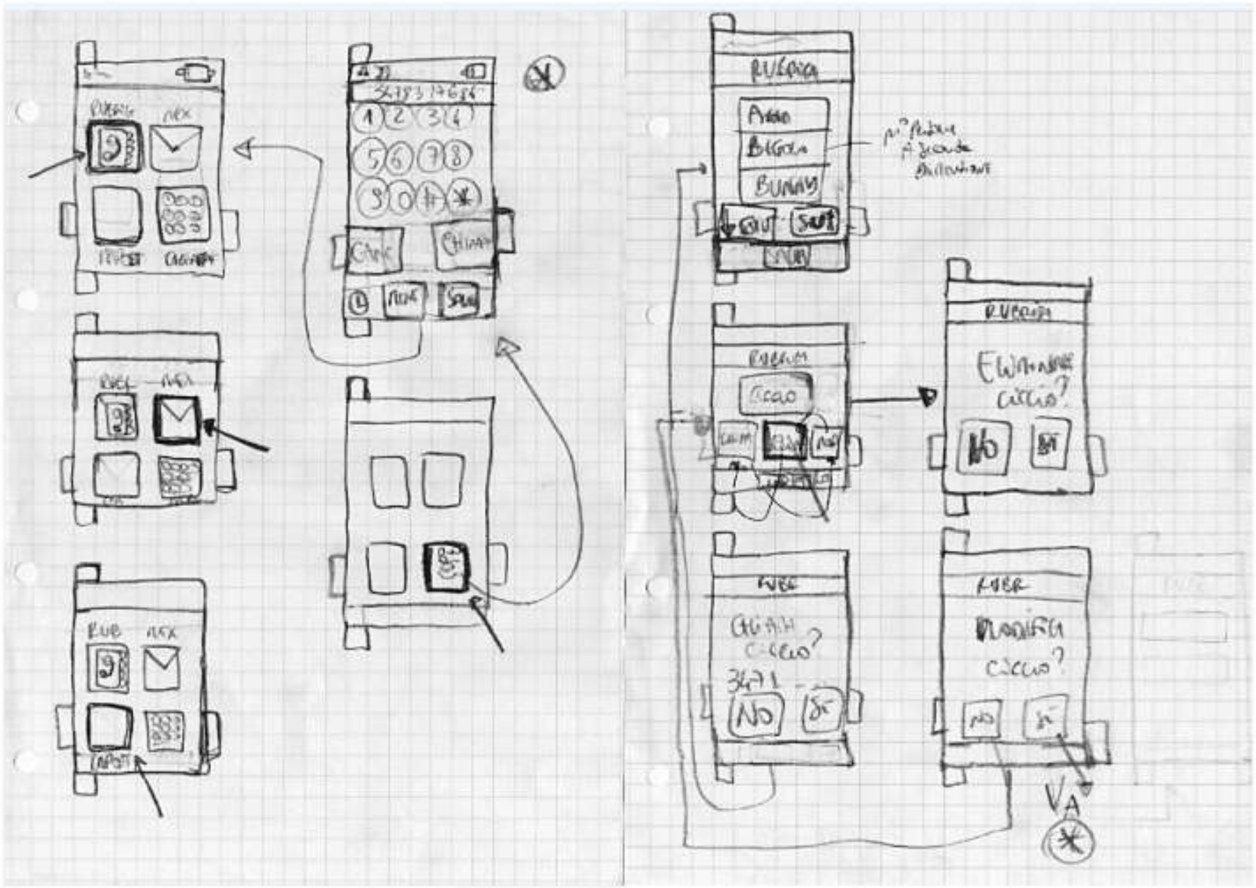
Prototypes

- Low fidelity
  - Paper / pencil, sketches, post-its
- High fidelity
  - Computer executable mock ups
    - Aka Powerpoint
    - Aka Balsamiq, Figma, ..
  - Actual GUIS
    - GUI Builders:
      - WindowBuilder (Eclipse, Java)
      - NetBeans Gui Builder (Net Beans)

Sketch

CLIENTE
SCHEDA
PAGATO
FATTURA
VIAGGI
LOCATION — SCHEDA
MEZZI — SCHEDA
ACCOMPAGNATORI — SCHED
CALENDARIO

Sketch / storyboard



Sketch

Feedback – low fi prototype

- Cognitive / ergonomy experts apply checklists / experience to identify possible issues

Feedback – hi fi prototype

- Selected users use the prototype in a lab
- Tools for Hi Fi prototypes
  - Balsamiq
  - Figma
- Feedback via
  - Ethnographics
  - Interviews
  - Focus group

Ethnographics

- End users perform their usual activities, using prototype
- Researcher(s) observe(s) the behaviour of end users, and especially interaction with prototype
  - Researcher tries to be invisible as much as possible to avoid modify behaviour of users

Interviews

- End users use prototype
- After use, (a subset of users) is interviewed
  - Interview = set of predefined questions, with open or closed answers, about the prototype

  - Ex: questionnaire at end of Polito course

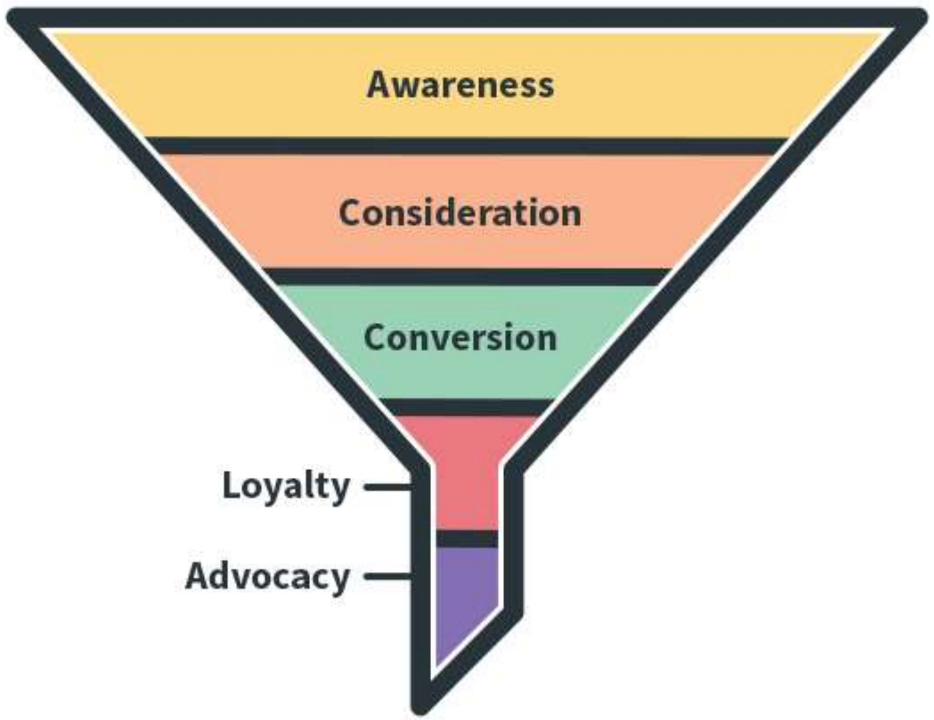
Focus group

- Users use prototype
- After use, they gather in a meeting, lead by a moderator
- Moderator invites users to express their opinions, ideas, suggestions
- Scribe takes notes, moderator draws conclusions

Feedback, final system

- Define and collect measures about
  - Usage of system (time spent on different pages / part of pages, errors)
  - Effect of system, conversion rate (ex rate from browse to purchase)

Marketing funnel



Conversion rate

- In general
  - From awareness of product / service
  - To customers purchasing
- On ecommerce website
  - From number of landing on web page per day by unique visitors
  - To number of purchases

Feedback

- A/B test

Designing the GUI

- Technical elements
- Usability guidelines

Technical elements

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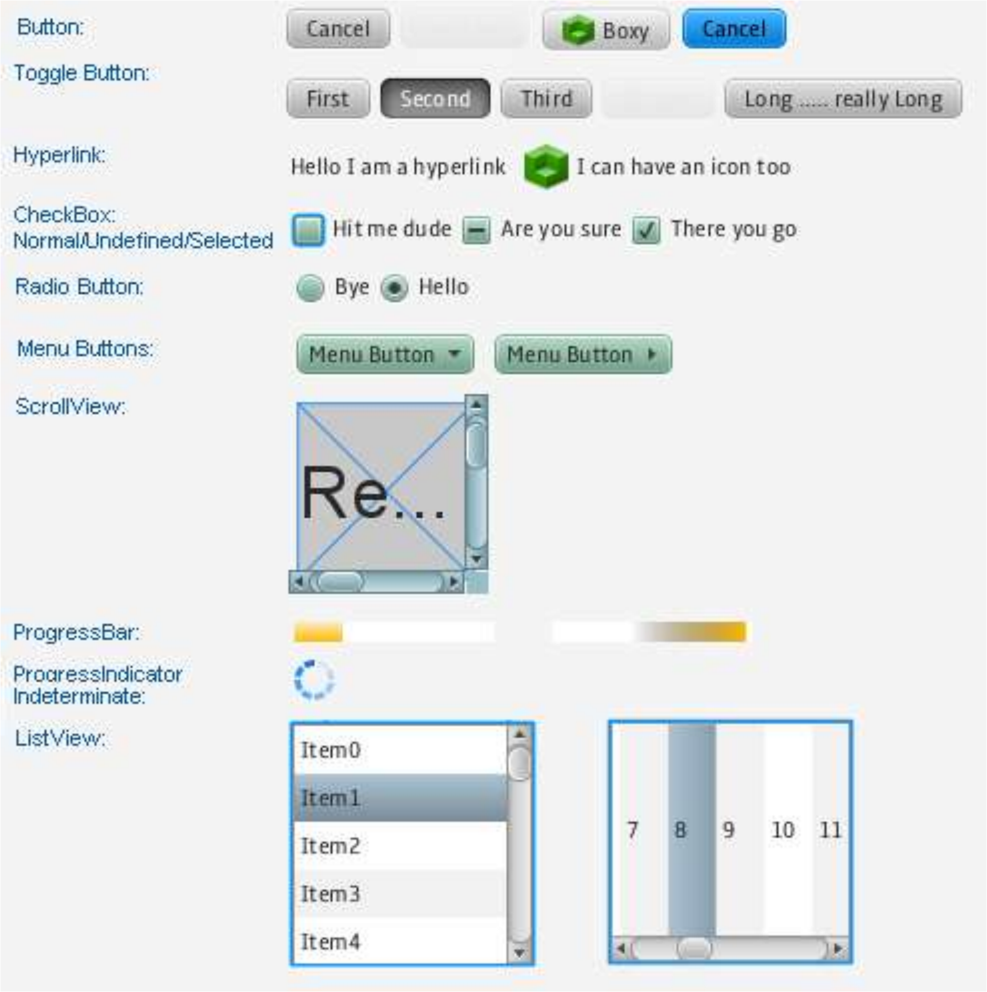
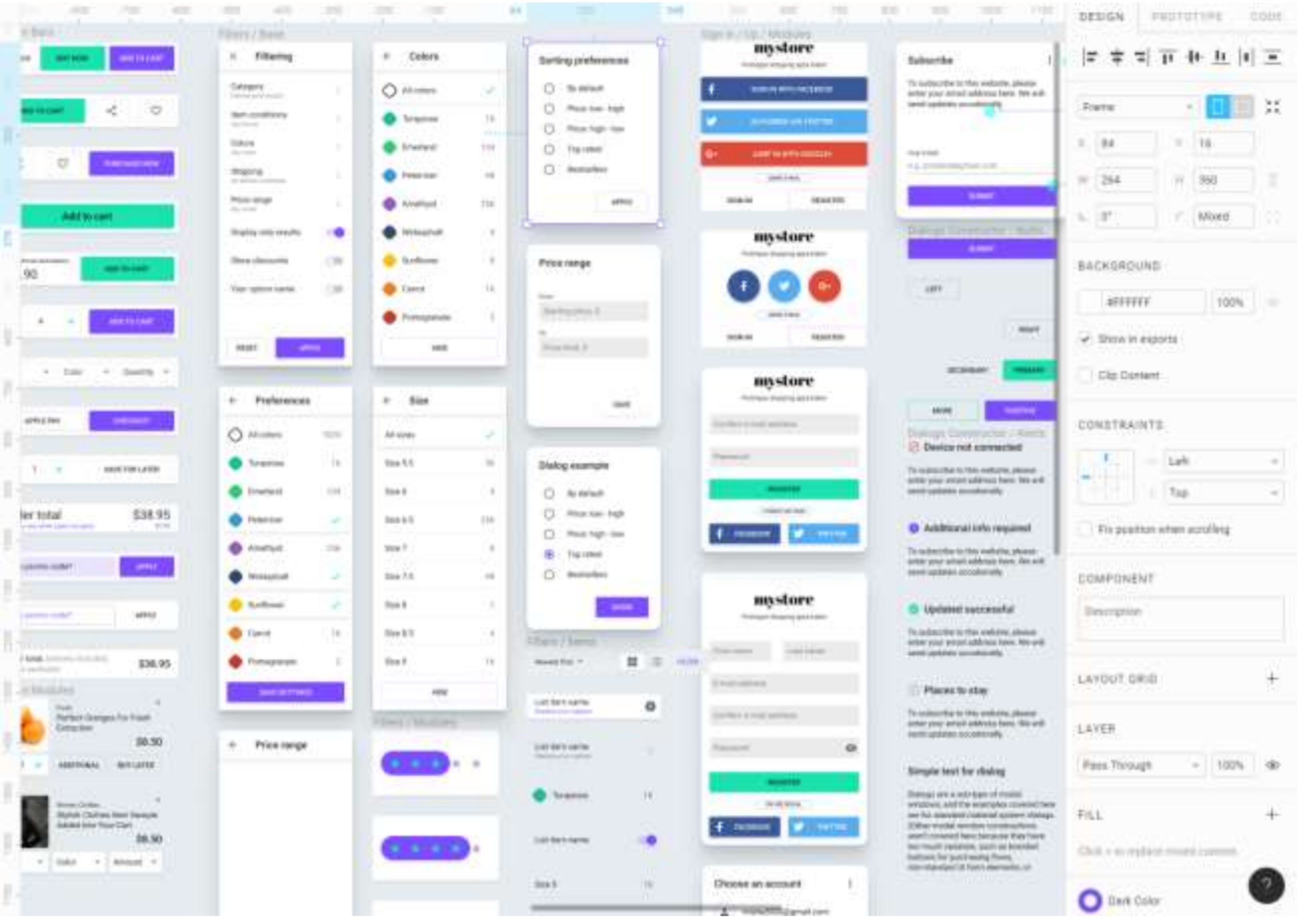
Java



Button:
Cancel
Boxy
Cancel
Toggle Button:
First
Second
Third
Long ...... really Long
Hyperlink:
Hello I am a hyperlink
I can have an icon too
CheckBox:
Normal/Undefined/Selected
Hit me dude
Are you sure
There you go
Radio Button:
Bye
Hello
Menu Buttons:
Menu Button
Menu Button
ScrollView:
Re...
ProgressBar:
ProgressIndicator
Indeterminate:
ListView:
Item0
Item1
Item2
Item3
Item4
7
8
9
10
11

Android



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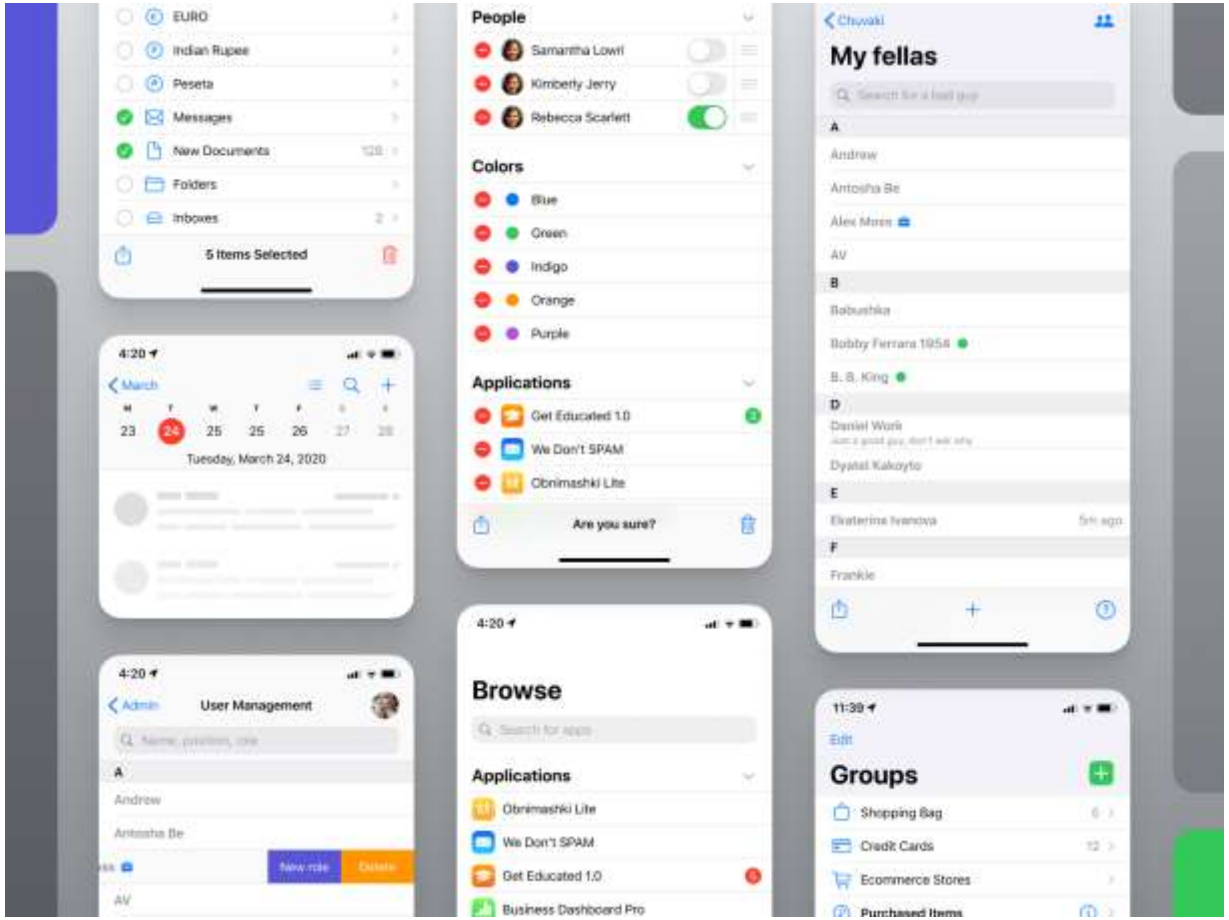
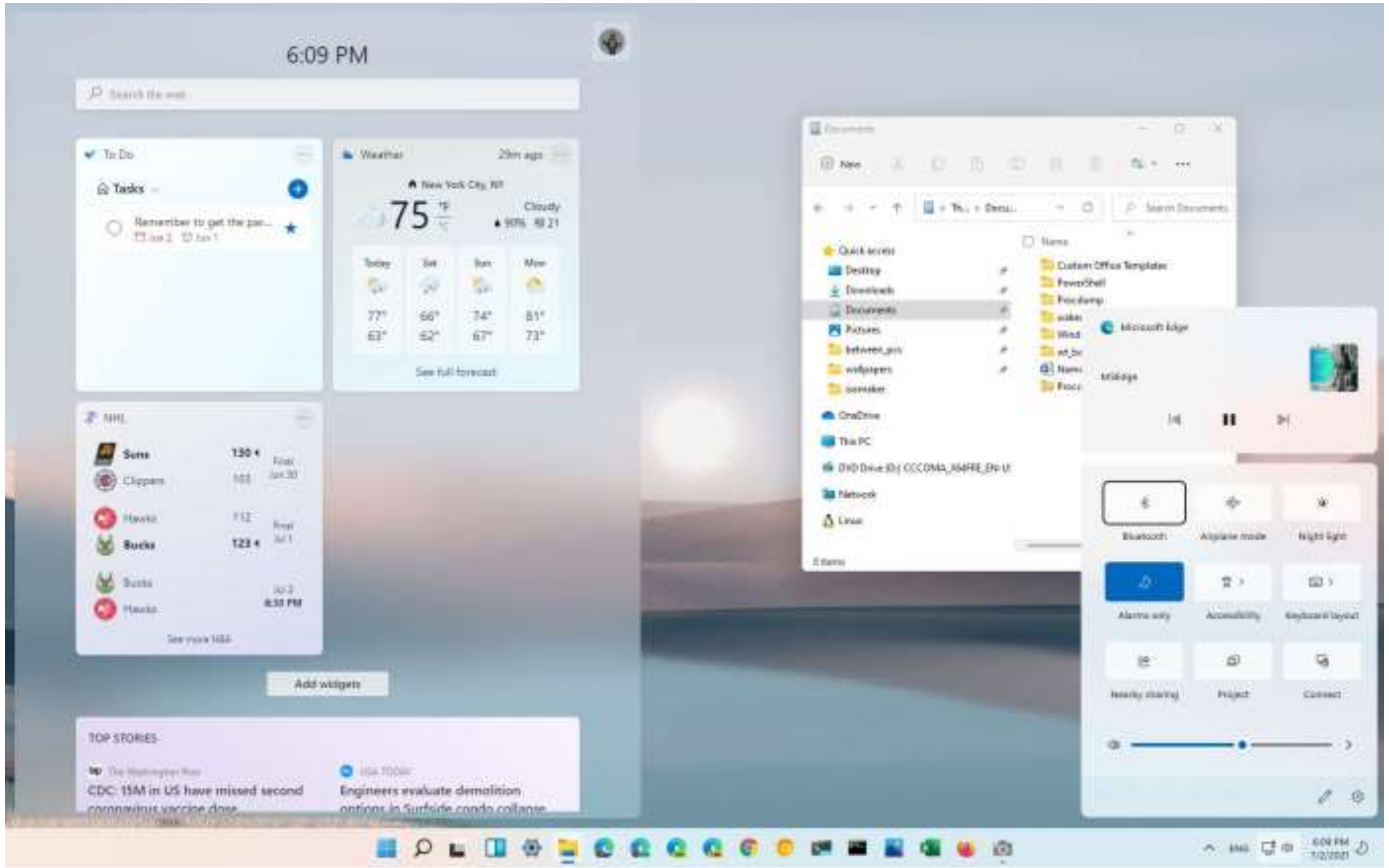
iOS


EURO
Indian Rupee
Peseta
Messages
New Documents
Folders
Inboxes
5 Items Selected
People
Samantha Lowri
Kimberly Jerry
Rebecca Scarlett
Colors
Blue
Green
Indigo
Orange
Purple
Applications
Get Educated 1.0
We Don't SPAM
Obnimashki Lite
Are you sure?
My fellas
Andrew
Antosha Be
Alex Moss
AV
Babushka
Bobby Ferrara 1958
B. B. King
Daniel Work
Dyatel Kakoyto
Ekaterina Ivanova
Frankie
March
Tuesday, March 24, 2020
User Management
Admin
Browse
Obnimashki Lite
We Don't SPAM
Get Educated 1.0
Business Dashboard Pro
Edit
Groups
Shopping Bag
Credit Cards
Ecommerce Stores
Purchased Items

Windows


6:09 PM
To Do
Tasks
Weather
75
See full forecast
NHL
Add widgets
TOP STORIES
CDC: 15M in US have missed second
Engineers evaluate demolition
Bluetooth
Airplane mode
Night light
Accessibility
Project
Connect

Issue

- Portability of the GUI

  - Redevelop GUI for each environment
  - Cross platform compilers
    - Ex Xamarin, Cordova, Flutter, ..
  - Cross platform GUI
    - Browser

Usability guidelines

- Have same style and format in all pages
- Do not ask same info twice
- Give feedback
  - When button clicked, when text inserted, when processing
- Make interactive objects obvious
  - Larger buttons, blinking, ..

Usability guidelines

- Consider default values in input fields
- Clear success / error messages
- Show clearly navigation hierarchy
  - Use breadcrump trails

Usability guidelines

- Simplicity /readability
  - Min number of pages
  - Min number of colors / fonts
  - Font min size
  - N elements in page
  - N pages
- Use conventions
  - Logo at top left
  - Click on logo brings to home
  - Links change color when mouse hovers
  - Next / back always in the same place

Summary

- In mass market products User interaction is key
- User centered design
  - Focuses on user feedback
  - Using several techniques in a defined process
- Key message: UI design must be validated with users
  - Never assume you did it right